

ent environment. Engineers can use intelligent thermal designs to prevent heat-related failures, increase the life expectancy of the systems and reduce emitted noise, energy consumption, cost and time-to-market.

Heat conduction occurs via motion of electrons and phonons. While the bulk properties of the materials determine the fundamental transport and scattering of these heat carriers, properties of interfaces and finite dimensionality are key issues in applications.

Three driving forces are placing stringent demands on materials for thermal management.

First, the drive to improve speed motivates circuit designers to compress the core of the microprocessor, which contains the region of the most electrical activity, to ever smaller sizes. This gives higher rates of heat generation per unit area. As a result, temperature drop caused by solid conduction within the chip and the heat spreader is predicted to become comparable to the maximum allowed temperature drop. The problem is compounded by exploratory three-dimensional circuits, which increase connectivity but also increase the difficulties associated with heat removal.

Second, temperature rise in interconnects between transistors is growing. This is due to increases in the number of metal layers and increasing interconnect current densities and aspect ratios. The problem is made worse by planned transition to interconnect passivation with lower dielectric constants than silicon dioxide (SiO_2).

Finally, there is the increasing temperature rise within contemporary and planned transistor technologies. Decreasing channel dimensions increases power density and electron-phonon non-equilibrium within devices. Novel device geometries (silicon-non-insulator, nano-pillars and fin-shaped field-effect transistors) offer greater thermal resistances to the bulk substrate.

These trends pose important challenges and opportunities for fundamental research in materials and solid-state physics.

Cooling of electronics

The rate of loss of heat from a hot object is governed approximately by Newton's

Law of Cooling. The law states that the rate of loss of heat is proportional to the temperature difference between the body and the surroundings. As temperature of the component rises, heat loss increases. When heat loss per second equates the heat produced per second within the component, the device achieves its equilibrium temperature. This temperature may be high enough to significantly shorten the life of the component, or even cause the device to fail. It is in such cases that thermal management measures need to be taken.

The same considerations can be applied to a complete circuit or device that incorporates heat-producing individual components. Rate of loss



Fig. 2: Use of heat-sink
(Credit: <https://en.wikipedia.org>)

of heat is higher in a forced draught than still air. One way of controlling the temperature of a device or circuit is to incorporate a fan, or fans, to increase air flow. Even ensuring adequate general ventilation results in a lower operating temperature than if the circuit is in a confined space with no ventilation slots.

One point that can be overlooked is that, reduced atmospheric density at high altitudes leads to less effective heat transfer to the surroundings and consequent higher device operating temperatures. There are several techniques for cooling, including various styles of heat-sinks, thermoelectric coolers, forced air systems and fans, heat pipes and so on. In cases of extreme low environmental temperatures, it may be necessary to heat the electronic components to achieve satisfactory operation.

Heat-sinks. Heat is lost from a component to its surroundings at the surface of the component. Rate of loss of heat increases with the surface

area of the component. A small device producing 10 watts reaches a higher temperature than a similar powered device with a larger surface area.

One way of limiting the operating temperature is to artificially increase the surface area. This is done by attaching a metal heat-sink to the device. A heat-sink can be made by stamping, extrusion or casting. It is generally fabricated from copper or aluminium or their alloys—the heat-sink needs to be a good conductor of heat.

Often, heat-sinks have a finned structure to maximise the available area for heat dissipation to the surroundings. Where heat-sinks are being used, these are more effective if the whole unit is well ventilated or, even better, forced air flow is applied by use of one or more fans.

It is impossible to make heat-sinks or components with perfectly flat mating surfaces, so when the two come into contact, high spots contact one another and there is a small air gap between the two over a large part of the surface.

Air is a poor conductor of heat and so the interface provides a thermal barrier. This limits the efficiency of heat loss from the device. It is to overcome this effect that heat transfer compounds are used.

More recently, synthetic diamond cooling-sinks are being researched to provide better cooling. Also, some heat-sinks are constructed of multiple materials with desirable characteristics, such as phase-change materials, which can store a great deal of energy due to their heat of fusion.

Heat spreaders. Another often-used cooling device is a heat spreader. It is a thermally conductive metal plate or foil used to move concentrated heat and distribute it over a wider area. It is normally used as an intermediate thermal interface between a heat source and a secondary heat exchanger, such as a heat-sink.

A heat spreader is often applied as a backing plate to PCBs containing heat-generating components. Thermal vias are placed in the PCB, which act as thermal conduits between each component package and the heat spreader to improve heat flow. Some